Exp: 13
1/10/26

Implementation of Ping Program

Aim:

To implement a simple ping program to check if a destination host is reachable and to measure the Round Trip Time of Packets.

Algorithm:

i) Start the program, and read the destination host name or IP address from the user.

ii) Create a raw ICMP socket and Echo request containing ICMP type, Code, Checksum, Identifier, Sequence number, Data.

iii) Calculate the checksum of the packet and send the ICMP echo request to the destination.

iv) Wait for a reply, then calculate the RTT.

v) Display the destination, packet size, sequence number, and RTT. Repeat this process for a specified number of packets.

Code:

```
import socket
import struct
import time
import os

def checksum(data):
    total = 0
    length = len(data)

    for i in range(0, length - 1, 2):
        total += (data[i] << 8) + data[i + 1]

    if length % 2:
        total += data[-1] << 8

    total = (total >> 16) + (total & 0xffff)
    total += total >> 16

    return (~total) & 0xffff


def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
    except socket.gaierror:
        print("Unable to resolve host")
        return

    icmp = socket.getprotobyname("icmp")

    try:
        sock = socket.socket(socket.AF_INET, socket.SOCK_RAW, icmp)
    except PermissionError:
        print("Permission denied. Run the program with administrator/root privileges.")
        return

    sock.settimeout(2)

    packet_id = os.getpid() & 0xFFFF
    sequence = 1

    # ICMP Echo Request
    header = struct.pack("!BBHHH", 8, 0, 0, packet_id, sequence)

    data = b"Computer Networks Ping"
    packet = header + data

    # Calculate checksum
    chksum = checksum(packet)

    # Re-create header with checksum
    header = struct.pack("!BBHHH", 8, 0, chksum,
                         packet_id, sequence)

    packet = header + data

    start_time = time.time()

    try:
        sock.sendto(packet, (dest_addr, 1))
        sock.recvfrom(1024)

        end_time = time.time()
        rtt = (end_time - start_time) * 1000

        print("Reply from", dest_addr,
              ": bytes =", len(packet),
              "time = %.2f ms" % rtt)

    except socket.timeout:
        print("Request timed out")

    finally:
        sock.close()


# Main program
host = input("Enter hostname or IP address: ")

for i in range(4):
    ping(host)
    time.sleep(1)
```

Outp

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devyes

Reply
Reply
Reply
Reply
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Output :

```
devyesh@Macs-MacBook-Air Python % Enter hostname or IP address: 8.8.8.8

Reply from 8.8.8.8 : bytes = 43 time = 24.51 ms
Reply from 8.8.8.8 : bytes = 43 time = 23.87 ms
Reply from 8.8.8.8 : bytes = 43 time = 24.12 ms
Reply from 8.8.8.8 : bytes = 43 time = 24.36 ms
```

DVP ✓

Result :

The implementation of Ping Program has been completed successfully.